STA 4001 Homework 3

1

Let's assume that the process starts at i, i.e. $X_0 = i$, i can be any possible random state in S. Since the Markov chain is irreducible, for $\forall j \in S$ and $j \neq i$, there exists a relative value n s.t. $P_{i,j}^n > 0$; since the Markov chain is recurrent, we can be certain that it will reenter i after time n.

Let's say the first opportunity is a success if $X_n = j$, and note that the 1st opportunity is a success with probability $P_{i,j}^n$. If the 1st opportunity fails then we consider the next time (after n) the chain enters i. Say the 2nd opportunity is a success if n time later the Markov chain is in j. Continuing in this manner, we can define an unlimited number of opportunities, each with success probability $P_{i,j}^n$. Since the number of opportunities until the first success occurs is geometric with parameter $P_{i,j}^n$, it will finally reach a success with probability 1, i.e. the chain will visit j with probability 1 at some future time.

Since i and j are randomly picked and $i \neq j$, starting from any state, the chain will visit every other state with probability 1 at some future time.

#2

(a)

Every day, the inventory level can remain the same, increase 1, or decrease 1, so for $\forall i, j \in \mathbb{N}$, there exists a positive integer n, s.t. $P_{ij}^{n} > 0$, so $\{X_n, n \geq 0\}$ is an irreducible DTMC.

$$P = \begin{bmatrix} (1-\lambda)+\lambda\mu & \lambda(1-\mu) & 0 & 0 & \cdots \\ (1-\lambda)\mu & (1-\lambda)(1-\mu)+\lambda\mu & \lambda(1-\mu) & 0 & \cdots \\ 0 & (1-\lambda)\mu & (1-\lambda)(1-\mu)+\lambda\mu & \lambda(1-\mu) & \cdots \\ 0 & 0 & \ddots & \ddots & \\ \vdots & \vdots & \ddots & & \end{bmatrix}$$

$$P(i,j) = \begin{cases} (1-\lambda)+\lambda\mu, & \text{if } i=0, j=0 \\ \lambda(1-\mu), & \text{if } i=0, j=1 \\ 0, & \text{if } i=0, j\geq 2 \\ (1-\lambda)\mu, & \text{if } i\geq 1, j=i-1 \\ (1-\lambda)(1-\mu)+\lambda\mu, & \text{if } i\geq 1, j=i \\ \lambda(1-\mu), & \text{if } i\geq 1, j=i+1 \\ 0, & \text{o.w.} \end{cases}$$

(b)

From 0, we must go to 1 on the very next jump. From any $i \geq 1$, we have positive probability of going to $i+1$ or $i-1$. Since from 1 we can go back to 0, and from 0 we can go to 1 with probability 1, all states communicate. Hence $\{Y_m\}$ is an irreducible DTMC.

$$P(Y_{m+1}=1 \mid Y_m=0)=1$$

Let's consider case $i \geq 1$, within a single day, the probability of $+1$ is $\lambda(1-\mu)$, the probability of -1 is $(1-\lambda)\mu$

$$P(X_{m+1}=X_m+1 \mid X_{m+1} \neq X_m)=\frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu}=p$$

$$P(X_{m+1}=X_m-1 \mid X_{m+1} \neq X_m)=\frac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu}=q$$

$$P(Y_{m+1}=i+1 \mid Y_m=i)=p,\quad P(Y_{m+1}=i-1 \mid Y_m=i)=q$$

Therefore, the transition matrix is:

$$P=\begin{bmatrix} 0 & 1 & 0 & 0 & \cdots \\ p & 0 & q & 0 & \cdots \\ 0 & p & 0 & q & \cdots \\ 0 & 0 & p & 0 & \cdots \\ \vdots & & & & \ddots \end{bmatrix}$$

where $\begin{cases} p=\dfrac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu} \\ q=\dfrac{(1-\lambda)\mu}{\lambda(1-\mu)+(1-\lambda)\mu} \end{cases}$

(c)

Observe that once the chain leaves 0, it then behaves like a simple "gambler's ruin" random walk on $\{1,2,3,\ldots\}$ with probability p of stepping up and $q=1-p$ of stepping down

A standard random walk on $\{0,1,2,\ldots\}$ with reflection at 0 satisfies:

- if $p > 1/2$, the chain drifts upward and is transient
- if $p = 1/2$, the chain is null recurrent
- if $p < 1/2$, the chain is positive recurrent

Transient: $p = \frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu} > \frac{1}{2} \Rightarrow \lambda > \mu$

Recurrent: $p = \frac{\lambda(1-\mu)}{\lambda(1-\mu)+(1-\lambda)\mu} \leq \frac{1}{2} \Rightarrow \lambda \leq \mu$

(d)

Since $\{X_n\}$ is a birth-death chain with some probability of "stay in place". Including a "stay put" probability does not affect whether it tends to drift up or down, so one gets exactly the same condition.

Transient: $\lambda > \mu$

Recurrent: $\lambda \leq \mu$

(e)

When $\lambda < \mu$, $\{X_n\}$ is a positive recurrent birth-death chain on $\{0, 1, 2, \cdots\}$. In that case it has unique stationary distribution $\{\pi_i\}_{i=0}^{\infty}$. We have:

$$\frac{\pi_{i+1}}{\pi_i} = \frac{(\text{up probability from } i)}{(\text{down probability from } i+1)} = \frac{\lambda(1-\mu)}{(1-\lambda)\mu} = \rho$$

$$\Rightarrow \pi_i = \pi_0 \rho^i, \quad i \geq 0$$

Since $\sum_{i=0}^{\infty} \pi_i = 1$, we have

$\sum_{i=0}^{\infty} \pi_i$ converge if $p<1$ $\longrightarrow$ $\pi_0 \frac{1}{1-p} = 1$

$\Rightarrow \pi_0 = 1-p$, $\pi_i = (1-p)p^i$, for $i = 0, 1, 2, \cdots$

where $p = \frac{\lambda(1-\mu)}{(1-\lambda)\mu} < 1$, so $\lambda < \mu$

Therefore, in the long run, $\{X_n\}$ spends a proportion π_i of time in state i, where

$$\pi_i = (1-p)p^i, \quad p = \frac{\lambda(1-\mu)}{(1-\lambda)\mu}, \quad \lambda < \mu$$

#3

(a)

Consider state i, let's denote $f(j)$ as the probability of hits i before hitting 0, starting from j.

$$f(0) = q_i p f(1) + q_i(1-p)f(n) + (1-q_i)f(0)$$

$$\Rightarrow f(0) = p f(1) + (1-p) f(n) \quad (*)$$

On the circle $\{0, 1, \cdots, n\}$, declare 0 and i are absorbing state, we have: $f(0) = 0$, $f(i) = 1$

Similar to $(*)$, we have

$$f(i) = p f((i-1) \bmod (n+1)) + (1-p) f((i+1) \bmod (n+1))$$

$$\Rightarrow \quad p\left[f(i) - f((i-1) \bmod (n+1))\right] = (1-p)\left[f((i+1) \bmod (n+1)) - f(i)\right]$$

$$\Rightarrow \quad \frac{f(i) - f((i-1) \bmod (n+1))}{f((i+1) \bmod (n+1)) - f(i)} = \frac{1-p}{p} = r$$

If $0 < x < i$, then $f(x) = \dfrac{\sum_{k=0}^{x-1} r^k}{\sum_{k=0}^{i-1} r^k}$, $r = \dfrac{1-p}{p}$

If $i < x \leq n$, then $f(x) = \dfrac{\sum_{k=x}^{n} r^k}{\sum_{k=i}^{n} r^k}$, $r = \dfrac{1-p}{p}$

(b)

$$p^n + (1-p)^n$$

(c)

Observe that if you do indeed visit all of $\{1, \cdots, n\}$ before returning to 0, you must have made a single "sweep" around the circle, either clockwise or counterclockwise - possibly with some back-and-forth among the inside stations, but never crossing back 0.

Since the problem requires $T_1 < T_2 < \cdots < T_n$, the only possibility is a clockwise "sweep".

$$P(T_1 < T_2 < \cdots < T_n \mid \text{all visited before return to } 0) = \frac{p^n}{p^n + (1-p)^n}$$

#4

$$E(X_i) = n\mu^i$$

$$E\left(\sum_{i=0}^{\infty} X_i\right) = \sum_{i=0}^{\infty} E(X_i) = n\sum_{i=0}^{\infty} \mu^i = \frac{n}{1-\mu}$$